

2nd Project - Machine Learning

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Final Deliverables:

1. Slide deck PDF pushed to GitHub designed for non-technical stakeholders that outline findings, recommendations, and future work (10 min presentation).
2. Jupyter notebook following PEP8 designed for data science/technical audience.
3. Optional: Python script for generating (takes .csv as argument and saves the model locally) and running your model from the terminal (takes test.csv and model as arguments and outputs accuracy and predictions as .csv) (you can still use the template repo: <https://github.com/neuefische/ds-ml-project-template>)
4. 1 slide about a potential Data product - How the predictions can be used?

Things to think about:

- Check for data imbalance!
- What would be the right performance metric - precision, recall, accuracy, F1 score, or something else? (Check TPR?)
- Try different (at least 3) machine learning algorithms to check which performs best on the problem at hand (including cross validation and hyperparameter tuning)

Submission Date:

- Friday, June 26th, 2026

1. Kickstarter Project Success (Medium)

In recent years, the range of funding options for projects created by individuals and small companies has expanded considerably. In addition to savings, bank loans, friends & family funding and other traditional options, crowdfunding has become a popular and readily available alternative.

Kickstarter, founded in 2009, is one particularly well-known and popular crowdfunding platform. It has an all-or-nothing funding model, whereby a project is only funded if it meets its goal amount; otherwise no money is given by backers to a project.

A huge variety of factors contribute to the success or failure of a project—in general, and also on Kickstarter. Some of these are able to be quantified or categorized, which allows for the construction of a model to attempt to predict whether a project will succeed or not. The aim of this project is to construct such a model and also to analyse Kickstarter project data more generally, in order to help potential project creators assess whether or not Kickstarter is a good funding option for them, and what their chances of success are.

- [Kickstarter Projects](#) (3 people)

2. Zindi Challenges (Easy - Advanced)

Please choose one of the following challenges on Zindi. You need to follow the requirements from the challenge and also make a stakeholder presentation:

- [Tanzania Tourism Prediction](#) (Easy - 2 people)
- [Fraud Detection in Electricity and Gas Consumption Challenge](#) (Medium, with starter notebook - 2-3 people)
- [Learning Urban Air Pollution Challenge](#) (Medium/Advanced - 3 people and domain knowledge)
- [Flight Delay Prediction Challenge](#) (and [airport data](#)) (Medium - 3 people)
- [Financial Inclusion In Africa](#) (Easy - 2 people)

3. Other Projects

A. Zillow Prize: Zillow Home Value Prediction (Zestimate)

- **Company:** Zillow (real estate platform).
 - **Task:** Predict the **log error** between Zillow's Zestimate (automated house price estimate) and actual sale price.
 - **Target Variable:** Continuous value (**logerror**).
 - **Why good:** Tabular data with rich real estate features. Clear regression problem with strong business use case.
 - **Link:** [Zillow Prize 1](#)
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B. Mercedes-Benz Greener Manufacturing

- **Company:** Mercedes-Benz.
 - **Task:** Predict the **time a car takes on the test bench** (continuous target) based on configuration features.
 - **Target Variable:** Continuous value (**y** = testing time).
 - **Why good:** Structured, tabular dataset from a real automaker with a practical regression target.
 - **Link:** [Mercedes-Benz Greener Manufacturing](#)
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C. Santander Value Prediction Challenge

- **Company:** Santander Bank.
- **Task:** Predict a **continuous customer value score** from anonymized customer data.
- **Target Variable:** **target** (continuous).
- **Why good:** Pure regression problem from the finance sector, structured tabular format.

- **Link:** [Santander Value Prediction*](#)
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D. Allstate Claims Severity

- **Company:** Allstate Insurance.
 - **Task:** Predict the **cost of insurance claims** (continuous severity value).
 - **Target Variable:** **loss**.
 - **Why good:** Insurance industry dataset, structured tabular features, straightforward regression.
 - **Link:** [Allstate Claims Severity](#)
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E. Expedia Hotel Recommendations (Regression Variant)

- **Company:** Expedia.
 - **Task:** Predict the **likelihood of a user booking a hotel cluster**.
 - **Note:** Mostly framed as classification, but regression approaches (predicting probability or time to booking) are also standard.
 - **Why good:** E-commerce & travel industry dataset, large and structured.
 - **Link:** [Expedia Hotel Recommendations](#)
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F. Porto Seguro's Safe Driver Prediction

- **Company:** Porto Seguro (insurance).
- **Task:** Predict whether a driver will **file an insurance claim** in the next year.
- **Target Variable:** Binary (**target** = 0 or 1).
- **Why good:** Large insurance dataset, real business case, tabular only.

- **Link:** [Porto Seguro Safe Driver](#)
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G. Santander Customer Transaction Prediction

- **Company:** Santander Bank.
 - **Task:** Predict which customers will make a **specific transaction**.
 - **Target Variable:** Binary (**target**).
 - **Why good:** Pure classification, anonymized structured dataset, from a major bank.
 - **Link:** [Santander Transaction Prediction](#)
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H. Expedia Hotel Recommendations

- **Company:** Expedia (travel).
 - **Task:** Classify which **hotel cluster** a user is most likely to book.
 - **Target Variable:** Multi-class (**hotel_cluster**).
 - **Why good:** E-commerce/travel domain, real-world tabular dataset, strong classification framing.
 - **Link:** [Expedia Hotel Recommendations](#)
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I. Avito Demand Prediction (Russian e-commerce)

- **Company:** Avito (online classifieds, like Craigslist).
- **Task:** Predict if an ad will be **successful (demand)**.
- **Target Variable:** Classification labels (conversion).
- **Why good:** Business dataset from a large e-commerce company.
- **Link:** [Avito Demand Prediction](#)

J. Home Credit Default Risk

- **Company:** Home Credit Group (lending/fintech).
- **Task:** Predict whether a **loan applicant will default**.
- **Target Variable:** Binary (**TARGET** = 0 or 1).
- **Why good:** Rich structured dataset, common real-world ML use case in fintech.
- **Link:** [Home Credit Default Risk](#)